Brief Project Description

Project Title: AI-Powered Student Placement Performance Analytics & Dashboard Using Machine Learning

Overview

PlacementIQ is an AI-powered student placement analytics and dashboard platform designed to help students and faculty understand placement readiness using academic, technical, and behavioral data. The system transforms student information into actionable insights through machine learning, explainable AI, analytics, and an AI assistant.

AI & Machine Learning

The platform uses a **Random Forest** model to predict student placement readiness using features such as CGPA, attendance, technical skill score, aptitude, communication, projects, internships, certifications, coding assessment performance, and backlogs. **K-Means clustering** is used to group students according to performance patterns, while **SHAP** provides explainability by showing the factors that contribute to individual predictions.

Key Features

Student	Profile and academic analysis; placement readiness; recommendations; skill-gap insights.
Faculty	Student data management; performance and readiness analysis; reports; historical placement analysis.
Analytics	Dashboard summaries, performance analysis, cluster distribution, at-risk student identification, and natural-language an
AI Assistant	AI-assisted interaction with placement information and analytics through the backend AI assistant.

Technology Stack

Frontend: React.js **Backend:** Python FastAPI **Database:** PostgreSQL **ML/AI:** Python, Scikit-learn, SHAP and AI assistant integration.

Snapdragon-Oriented Deployment

The project is structured for local AI deployment and is intended to be optimized for Snapdragon-powered HP PCs. The architecture separates the application interface, backend services, machine-learning models, and AI assistant so that local inference can be integrated and evaluated on the target Snapdragon platform. Snapdragon-specific hardware acceleration is claimed only when verified through actual deployment and testing on the target device.

Expected Outcome

PlacementIQ provides a centralized, data-driven view of student placement performance, helping students identify areas for improvement and enabling faculty to analyze readiness, performance groups, skill gaps, and placement-related trends through an interactive AI-enabled platform.

Project Type: Educational Data Mining / Artificial Intelligence & Data Science